

Name: _____

Quiz 2

MAT 336 - Information Systems

17 September 2026

Refer to the additional sheet handed out with this quiz that shows screen shots of a web page of Amtrak train routes and the browser inspector view of part of the table shown on that web page. The first table row has its elements expanded. Assume all other table rows have the same structure.

Assume the following code has been run to load Python modules, download this Amtrak web page, and import it into a BeautifulSoup object:

```
import requests
from bs4 import BeautifulSoup
import pandas as pd

page = requests.get('https://en.wikipedia.org/wiki/List_of_Amtrak_routes')
soup = BeautifulSoup(page.content, 'html.parser')
```

Almost all of the questions build on the prior ones. Answer each one as if your previous answers are correct.

1. Write a Python statement that finds the table containing the current Amtrak routes and assigns it to a Python variable named `table`. Hint: it is the first table on the page.
2. Write a Python statement that finds all of the rows in `table` and assigns them to a variable named `rows`.
3. Write Python code to remove the table's header row from `rows` and store it in its own variable named `hdr_row`.
4. Write code to store the 1st table row (the Acela row) from `rows` in a Python variable named `row`.
5. Write a statement that finds all of the data items (or cells) in `row` and stores them in a Python variable `items`.

6. Write Python code that fills a new list named `item_text` with the `text` found within each of the items in `row` in the previous question.

7. Write code that finds any URL links found within the row and stores them in a Python list named `links`.

8. Assume the table headers have been stored in a list as follows:

```
hdrs = ['Name', 'Type', 'Route', 'Numbers', 'Daily round trips',  
        'FY2025 passengers', 'Route miles']
```

Write code that creates a Python dictionary named `row_dict` that combines `hdrs` with the text of the items in row 1 (the Acela row) found earlier.

9. The code up to and including the previous question scrapes one row of the table into a Python dictionary. Write in words and/or pseudo-code how you would scale this up to scrape all rows of the table, putting each row's content into a dictionary. You may refer to previous question numbers in your answer. Assume you do not need to capture any URLs.
10. Suppose you completed scraping all rows of the table on the web page into a list of Python dictionaries named `row_dicts_list`. Write a statement that creates a `pandas` DataFrame out of them.